Hands-on Activity 2.1	
Data Types and Arithmetic Operations	
Course Code: CPE 007	Program: Computer Engineering
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1. Objective(s)

This activity aims to develop programming skills in structuring data types and arithmetic operations.

2. Intended Learning Outcomes

The students should be able to:

2.1 Find possible compilation errors and logic errors

2.2 Complete programs applying data types and arithmetic operations

3. Materials and Equipment

Internet connection
Personal Computer

4. Procedure

Discussions:

Identifiers are names of variables, functions, and arrays. They are user-defined names, consisting sequence of letters and digits, with the letter as the first character.

Constants - values that do not change during the execution of program

Numerical Constants

— Integer constants - These are the sequence of numbers from 0 to 9 without decimal points or fractional part or any other symbols. It requires minimum two bytes and

maximum four bytes.

Eg: 10,20, + 30, – 14

— Real constants - It is also known as floating point constants.

Eg: 2.5, 5.342

Character Constants

— Single character constants - A character constant is a single character. Characters are also represented with a single digit or a single special symbol or white space enclosed within a pair of single quote marks

Eg: ‘a’, ‘8’, “ ”.

— String constants - String constants are sequence of characters enclosed within double quote marks.

Eg: “Hello”, “india”, “444”

Variables - It is a data name used for storing a data value. Its value may be changed during the program execution. The value of variables keeps on changing during the execution of a program.

table

Data type	Size (Bytes)	Range	Format Specifiers

Char	1	-128 to 127	%c

Unsigned char	1	0 to 255		%c

Short or int	2	– 32,768 to 32, 767	%i or %d

Unsigned int	2	0 to 655355	%u

Float	4	3.4e – 38 to +3.4e +38	%f or %g

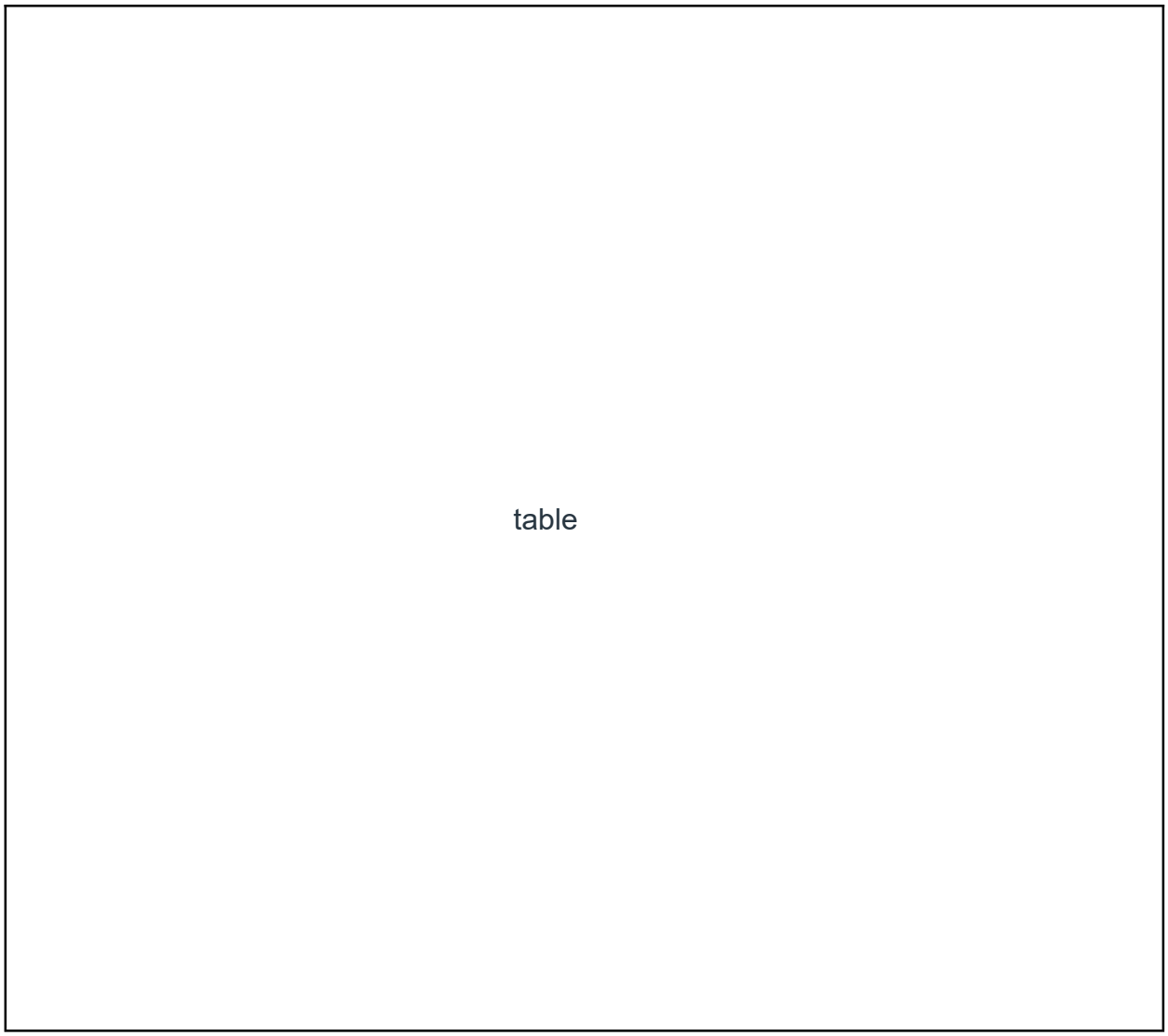
Long	4	= 2147483648 to 2147483647	%ld

Unsigned long	4	0 to 4294967295	%lu

Double	8	1.7e – 308 to 1.7e+308	%lf

Long double	10	3.4e – 4932 to 1.1e+4932	%lf

Operators - It indicates an operation to be performed on data that yields value.


table

Type of Operator	Symbolic representation

Arithmetic operators	+, -, *, /, %

Relational operators	>, <, ==, >=, <=, !=

Logical operators	&&, ||, !=

Increment and decrement operator	++ and --

Assignment operator	=

Bitwise operator	&, |, ^, >>, <<, ~

Comma operator ,

Conditional operator	?:

Input and Output

Reading data from input devices and displaying the results on the screen are the two main tasks of any program.

Formatted Functions

- The formatted input/output functions read and write all types of values
- Input = scanf
- Output = printf

Unformatted Functions

- The unformatted input/output functions only work with the character data type

- Input = getch(), getche(), getchar(), gets()
- Output = putch(), putchar(), put()

Example 1: The following program has an output of:

The value of seven is: 7.000000

The value of eight and a half is: 8.500000

Can you find all possible compilation errors and logic errors? Can you fix them to print the same result as the expected output? Before you use your compiler, try to find the errors only by manual code analysis.

```
#include<iostream>

using namespace std;

int main()

{

cout<<"The value of seven is: ";

cout<<"The value of eight and a half is: ", <<8.5;

return 0;

}
```

Example 2: The following program has an output of:

The value of seven is: 7.000000

The value of eight and a half is: 8.500000

Can you find all possible compilation errors and logic errors? Can you fix them to print the same result as the expected output? Before you use your compiler, try to find the errors only by manual code analysis.

```
#include <iostream>

using namespace std;

int main()

{

cout<<"The value of seven is: "<< 7 0;

cout<<"The value of eight and a half is: "<<8.5;
```

```
return 0;
}
```

Example 3: The following program has an output of:

The value of half is: 0.500000

The value of Pi is: 3.141593

Can you find all possible compilation errors and logic errors? Can you fix them to print the same result as the expected output? Before you use your compiler, try to find the errors only by manual code analysis.

```
int main()
{
float halfValue = 0.6;
float piValue = 3.141 592 65;
cout<<"The value of half is: "<< half Value;
cout<<"The value of Pi is: "<<pi_Value;
return 0;
}
```

Example 4: Sample program for Adding Two Integers

```
#include <iostream>
int main()
{
int integer1, integer2, sum; /*declaration */
cout<<“Enter first integer: \n” ;  /* prompt */
cin>>integer1 ;          /* read an integer */
cout<<“Enter second integer: \n” ; /* prompt */
cin<<integer2;          /* read an integer */
sum = integer1 + integer2;         /* assignment of sum */
cout<<“Sum is : "<<sum;       /* print sum */
```

return 0; / indicate that program ended successfully */*

}

Sample Output:

Enter first integer

45

Enter second integer

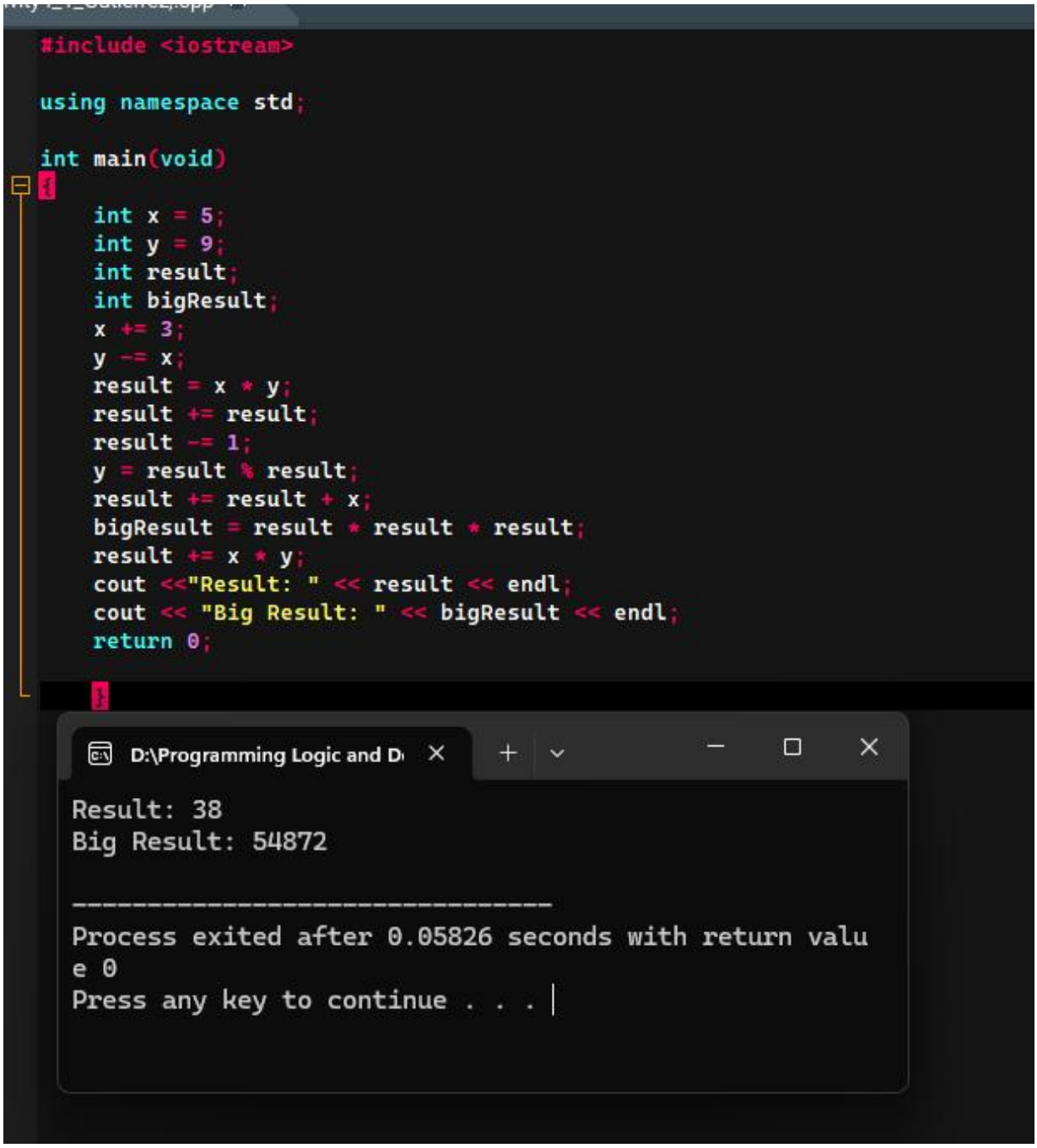
72

Sum is 117

5. Output

6. Additional Activity

1. Take a look at the code below: it assigns two integer values, manipulates them and finally outputs the result and bigresult variables. The problem is that the manipulations have been described using natural language, so the code is completely useless now. Act as an intelligent (naturally!) compiler and translate the formula into a real "C" code notation. Test your code using the data provided.



```cpp
#include <iostream>

using namespace std;

int main(void)
{
    int x = 5;
    int y = 9;
    int result;
    int bigResult;
    x += 3;
    y -= x;
    result = x * y;
    result += result;
    result -= 1;
    y = result % result;
    result += result + x;
    bigResult = result * result * result;
    result += x * y;
    cout <<"Result: " << result << endl;
    cout << "Big Result: " << bigResult << endl;
    return 0;

}
```

```
D:\Programming Logic and D
Result: 38
Big Result: 54872

--------------------------------
Process exited after 0.05826 seconds with return valu
e 0
Press any key to continue . . .
```

2. Complete the program below. Compute the accrued amount of money with a starting value of 100 and an annual interest rate of 1.5%. Compute and print the results for first three years. Your version of the program must print the same result as the expected output for every year. Compute each annual value on the basis of the previous year's value.

Activity4_2_Gutierrez.cpp

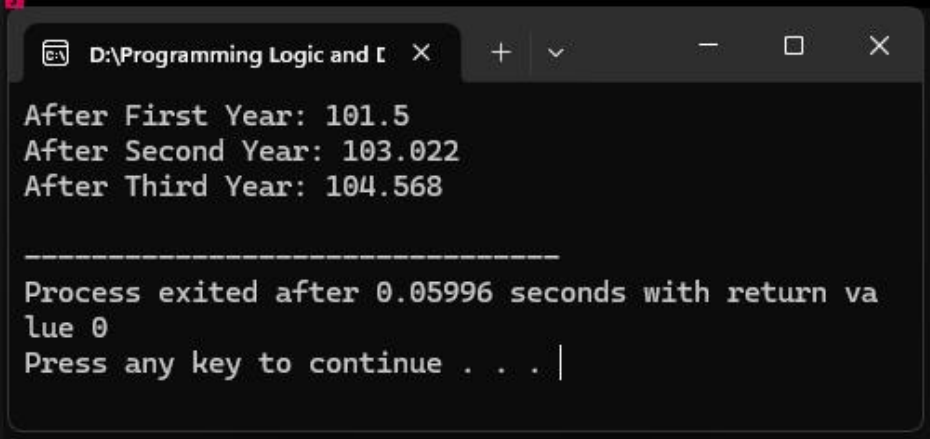
```cpp
#include <iostream>

using namespace std;

int main()
{
    float start = 100;
    float interestRate = 0.015;
    float Fy;
    float Sy;
    float Ty;
    Fy = start * (1+ interestRate );
    Sy = Fy * (1+ interestRate );
    Ty = Sy * (1+ interestRate );
    cout<<"After first year: "<<Fy<<endl;
    cout<<"After second year: "<<Sy<<endl;
    cout<<"After third year: "<<Ty;

return 0;

}
```



7. Conclusion

I have complicated programming in this lesson but I do not fully I understand it yet

8. Rubric